or anything has carried out a given observation—remains in the usual realm of quantum uncertainty, of probabilities. Even though we measure an electron's position as right here right now, a moment ago all it had were probabilities of being here, or there, or way over there.

And as we've seen, it is not that the electron (or any particle for that matter) really was located at only one of these possible positions, but we simply don't know which.[2] Rather, there is a sense in which the electron was at all of the locations, because each of the possibilities—each of the possible histories—contributes to what we now observe. Remember, we saw evidence of this in the experiment, described in Chapter 4, in which electrons were forced to pass through two slits. Classical physics, which relies on the commonly held belief that happenings have unique, conventional histories, would say that any electron that makes it to the detector screen went through either the left slit *or* the right slit. But this view of the past would lead us astray: it would predict the results illustrated in Figure 4.3a, which do not agree with what actually happens, as illustrated in Figure 4.3b. The observed interference pattern can be explained only by invoking an overlap between something that passes through *both* slits.

Quantum physics provides just such an explanation, but in doing so it drastically changes our stories of the past—our descriptions of how the particular things we observe came to be. According to quantum mechanics, each electron's probability wave *does* pass through both slits, and because the parts of the wave emerging from each slit commingle, the resulting probability profile manifests an interference pattern, and hence the electron landing positions do, too.

Compared with everyday experience, this description of the electron's past in terms of criss-crossing waves of probability is thoroughly unfamiliar. But, throwing caution to the wind, you might suggest taking this quantum mechanical description one step further, leading to a yet more bizarre-sounding possibility. Maybe each individual electron itself actually travels through both slits on its way to the screen, and the data result from an interference between these two classes of histories. That is, it's tempting to think of the waves emerging from the two slits as representing two possible histories for an individual electron—going through the left slit or going through the right slit—and since both waves